ZINGERZ TEAM 20

Tech Warmup II

Generative AI as an Engineering Tool

Research & Initial Strategy

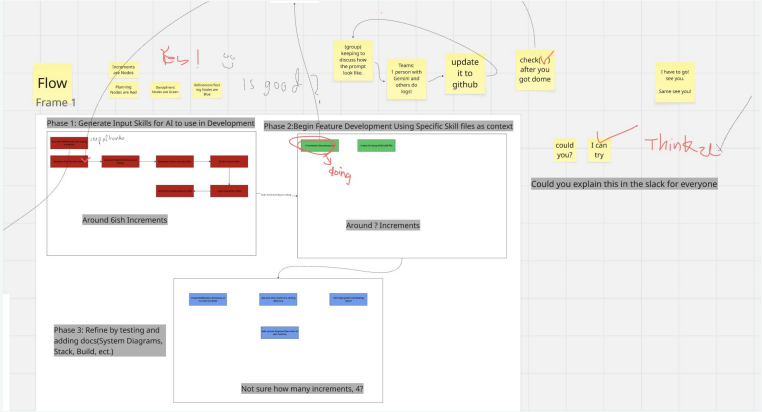
We built a modular 3-layer foundation, then used an increment plan to iteratively refine features, quality, and compliance outlined in the image below.

Agent & Model

Gemini

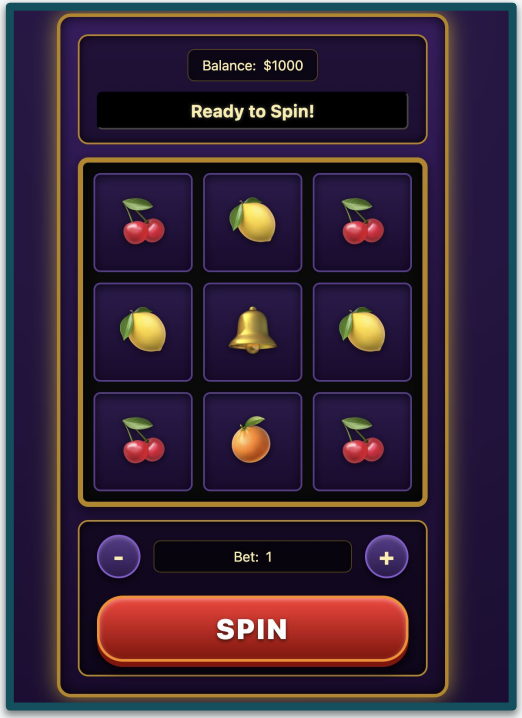
gemini-3.1-pro-preview

Each team member did research and adding their findings on miro.



Increments 2-5

- To ensure code quality:
  - Configured linting toolchain
    - ESLint, Prettier, stylelint, HTMLHint
  - Ran the full linter suite against the existing codebase and resolved all reported violations
- Refactored GameManager for easy testing without duplicate code (DRY)
- Implemented unit tests for the wallet and game controller

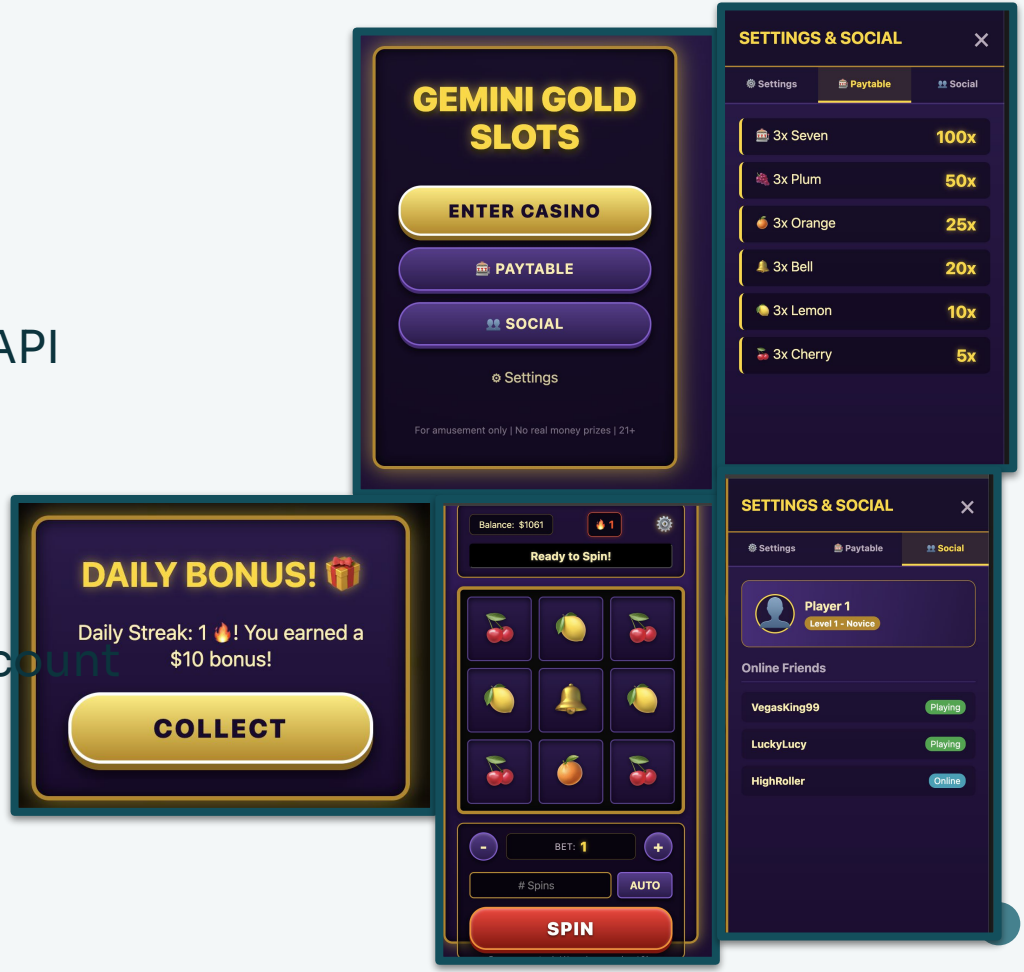


Increments 6–10

- Reel spin animation
  - Timing, deterministic outcomes, symbol cycling
- Intuitive win/loss feedback visuals
  - Highlight, audio, motion design queues
  - UI clarity & accessibility
- Legal compliance
  - Disclaimers on bottom for age, digital betting, entertainment
- Documentation cleanup
  - Feature, architecture, legal description

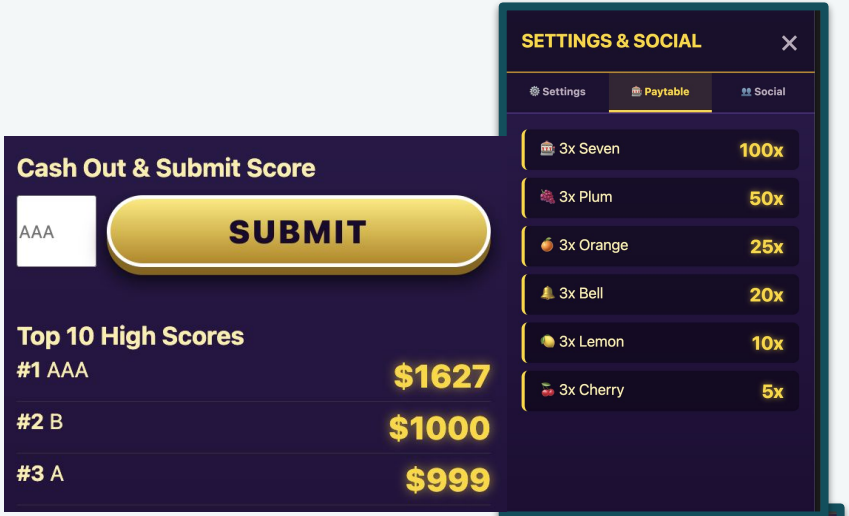
Increments 11-16

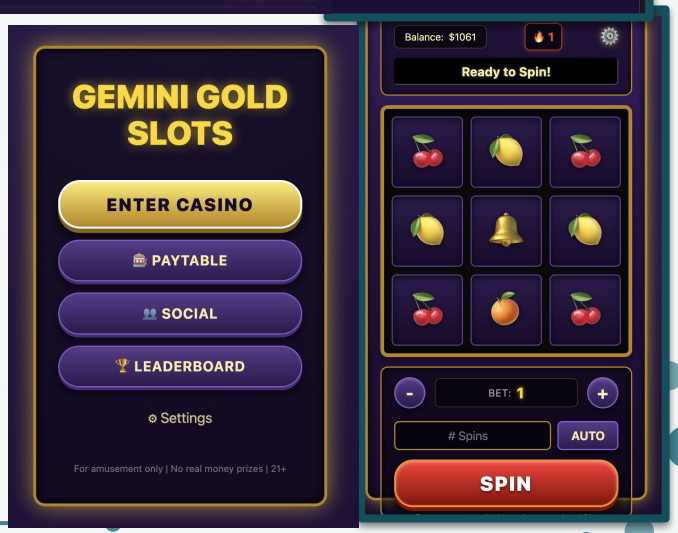
- Bets scale according to remaining balance
- All SFX through Web Audio API
  - No external audio files
- Main Menu, waiting to enter casino
- Social page
- Auto-Spin: mandatory spin count requirement
- Daily Streaks and Rewards



Increments 17-21

- Fixed ethical issues (paytable, Auto-spin, broke reward)
- Ran 500,000 simulated spins to check 96% RTP rate
- Full code quality check
- Full-Stack: Moved from local to centralized Node.js/Express architecture
  - Shared Leaderboard
  - Foundation for multiplayer play





Final Findings

Visuals were lackluster

Last project had better visuals but less features

Architecture and Security

Keeping layers separate paid off

Research actually mattered

Shaped a lot of design choices

Use of AI

AI helped accelerated development but still required review, refinement, and correction